

11.1

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Assume λ 's are real. Analyze stability and classify.

29. $A = \begin{bmatrix} 2 & -1 \\ 0 & 3 \end{bmatrix}$ triangular $\Rightarrow \lambda = 2, 3$

stability: unstable

classification: unstable node

$$\det(A - \lambda I) = 0$$

$$(2 - \lambda)(3 - \lambda) - (0)(-1) = 0$$

$$(2 - \lambda)(3 - \lambda) = 0$$

$$\lambda = 2, 3$$

$$\det(A - \lambda I) = 0$$

$$\lambda^2 - 5\lambda + 6 = 0$$

39. $A = \begin{bmatrix} 0 & -2 \\ -1 & 3 \end{bmatrix}$ $\Delta = (0)(3) - (-2)(-1) = -2 < 0$

stability: unstable

Classification: saddle node

$$\tau = 3$$

$$\lambda^2 - 3\lambda - 2 = 0$$

$$\lambda = \frac{3 \pm \sqrt{9 - 4(-2)}}{2} = \frac{3 \pm \sqrt{17}}{2}$$

opposite signs
same conclusion

$$3^2 (\sqrt{17})^2$$

$$9 < 17$$

$$3 - \sqrt{17} < 0$$

Assume λ 's are imaginary

$$\text{43. } A = \begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix} \quad \tau = 2$$

$$\Delta = (2)(0) - (-1)(3) = 3$$

Stability: unstable

Classification: unstable spiral

$$\text{51. } A = \begin{bmatrix} -1 & 1 \\ -3 & 1 \end{bmatrix} \quad \tau = (-1) + (1) = 0$$

Stability: ? (unstable)

Classification: center

$$\lambda^2 - 2\lambda + 3 = 0$$

$$\lambda = \frac{2 \pm \sqrt{2^2 - 4 \cdot 3}}{2}$$

$$= \frac{2 \pm \sqrt{4 - 12}}{2}$$

$$= \frac{2 \pm \sqrt{-8}}{2} = \frac{2 \pm i\sqrt{8}}{2}$$

$$= \frac{2 \pm 2i\sqrt{2}}{2}$$

$$= 1 \pm i\sqrt{2}$$

No assumptions on λ 's.

$$57. A = \begin{bmatrix} -1 & -2 \\ 1 & 3 \end{bmatrix}$$

$$\Delta = (-1)(3) - (1)(-2) = -3 + 2 = -1 < 0$$

saddle point ✓

(unstable)

$$\lambda^2 - \tau\lambda + \Delta = 0$$

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2}$$

$$\tau = (-1) + 3 = 2$$

$$\lambda = \frac{2 \pm \sqrt{2^2 - 4(-1)}}{2}$$

$$= \frac{2 \pm \sqrt{8}}{2} = \frac{2 \pm 2\sqrt{2}}{2}$$

$$= 1 \pm \sqrt{2}$$

$$1 - \sqrt{2} < 0$$

$$(1)^2 < (\sqrt{2})^2 \uparrow$$

$1 < 2$

$$61. A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \end{bmatrix}$$

$$\Delta = (1)(3) - (2)(3) = -3 < 0$$

saddle point

(unstable)

$$65. A = \begin{bmatrix} 3 & -5 \\ 2 & -1 \end{bmatrix}$$

$$\Delta = (3)(-1) - (2)(-5) = -3 + 10 = 7$$

$$\tau = 3 + (-1) = 2$$

unstable

$$\tau^2 - 4\Delta = (2)^2 - 4(7)$$

$$= 4 - 28 < 0$$

imaginary

unstable
spiral

$$64. A = \begin{bmatrix} -2 & -7 \\ 1 & 2 \end{bmatrix}$$

$$\Delta = (-2)(2) - (1)(-7)$$

$$= -4 + 7 = 3$$

$$\tau = 0$$

center

unstable node $\Leftrightarrow$ source

stable node $\Leftrightarrow$ sink

$$\tau^2 - 4\Delta = (0)^2 - 4(3) = -12 < 0$$